a population of more than 4 million with expansion to all state capitals and major highways connected to the mega cities in the following couple of years. Charging points can be installed as public, private, or for restricted public use.

Despite these challenges and lack of clarity in certain aspects of the regulation, public and private players alike are looking to bolster EV charging infrastructure to allow wider adoption and mitigate range anxiety. Tata Power, for instance, is putting up EV charging stations in malls and public access areas. Several dealerships are offering charging stations on their premises and service centers as well. EV manufacturers such as Tata Motors, Hyundai, MG, Mercedes Benz, Ather Energy, and others are now offering the ability to set up a charging station in your home, though things aren't often a smooth sailing with resident welfare associations.

Public enterprises such as National Thermal Power Corporation (NTPC) Limited, Power Grid Corporation of India Limited (PGCIL), Energy Efficiency Services Limited (EESL), and BEE together with Oil Marketing Companies (OMCs) are also involved in enhancing EV charging infrastructure. In fact, the OMCs have announced setting up 22,000 charging stations in all major cities and on national highways across India. Of these 22,000, 10,000 will be installed by the Indian Oil Corporation Limited (IOCL), 7,000 by Bharat Petroleum Corporation Limited (BPCL), and 5,000 by Hindustan Petroleum Corporation Limited (HPCL). The Department of Heavy Industry is also doing its part by sanctioning 1,576 public charging stations for 25 highways and expressways for every 25 km.

To address long charging times at EV stations, the Government of India is mulling on a Battery Swapping Policy. Battery swapping is already in vogue in a limited extent for smaller EVs such as two-wheelers, but the government plans to regulate this sector across all vehicles in order to save time and reduce costs. The draft battery swapping policy looks to standardize the type of batteries used in all EVs in the country for easy interoperability and to ensure mission-critical transport services are not affected. However, the pitfall here is to ensure that the batteries being swapped are of a certain standard with their regular wear and tear also being considered. That being said, battery swapping in four-wheelers is not an easy task given their increased sizes and complexity.

FINANCIAL CONSTRAINTS

India is an extremely price-conscious and price-sensitive market with budget two and four-wheelers traversing most Indian roads. Finances, both at the personal and at the industry level, play a decisive role in EV adoption. It is being estimated that India will have to spend upwards of US$65 billion per year to electrify its transportation. A major chunk of this expense goes towards the import of batteries, microcontrollers, motors, and other components that make up nearly 60% of a typical vehicle's cost.

Currently, the cost of purchasing an EV is much higher than a comparable ICE vehicle. Nearly 40-50% of this expense that an OEM incurs is surprisingly spent on the battery than other components. Raw materials that go into a battery including lithium, cobalt, nickel, and manganese often have to be imported since India has limited reserves of these ores. These components also require certifications such as an IP67 rating, which unfortunately doesn't seem to be quite standardized for EVs yet.

On an average, electric cars start from ₹13 lakh, which is more than twice the average cost of a budget ICE car. Electric scooters too have a higher upfront cost ranging from ₹70,000 to ₹1.25 lakh compared to the ₹30,000 to ₹40,000 cost for a comparable ICE two-wheeler. EVs also tend to have lesser configuration options and customizations to choose from compared to their ICE counterparts.

Therefore, the EV market currently does not benefit from the economies of scale even though there is a projected 43.13% CAGR for the EV industry from 2019 to 2030 and a 42.38% CAGR for development of charging infrastructure.

INITIATIVES TO MAKE EVS FINANCIALLY VIABLE

The Government of India has proposed several initiatives to help lower the cost of EV production and increase adoption.

The more famous among these initiatives is well, FAME, short for Faster Adoption and Manufacture of Hybrid and Electric Vehicles. FAME was first launched in 2015 to incentivize technology development, pilot new projects, and create the necessary charging infrastructure. The present FAME II scheme launched in 2019 with an outlay of 10,000 crores to support 500,000 electric three-wheelers, 7,000 electric buses,